

The quality of data & information

A data visualisation is only ever as trustworthy as the data underneath it. This module unpacks the five factors VCAA uses to judge that trust — accuracy, bias, integrity, relevance and reliability — and the sub-factors hiding inside one of them.

What this key knowledge covers

This summary distils everything you need for **KK03 — The quality of data & information** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

Quality data → quality information

The study design separates two words on purpose. **Data** is the raw, unprocessed facts Ava collects — 147, "basketball", TRUE. **Information** is what those facts *mean* once they're organised, analysed and presented — "most Year 9s want the courtyard for ball sports." The visualisation is information; the survey responses are data.

Here's the catch examiners love: **you cannot fix bad data with good presentation**. A beautiful chart built on rubbish numbers is a beautiful, convincing lie. Quality has to be protected at the data stage, before it ever becomes information.

Data is your ingredients; information is the finished dish. A brilliant chef cannot rescue a meal made from spoiled ingredients — they can only plate the spoilage attractively. Programmers say the same thing more bluntly: **garbage in, garbage out (GIGO)**.

VCAA assesses quality against five named factors. They overlap a little in real life, but in an exam you're expected to name them, define them, and apply them to a scenario — so we'll treat each one precisely.

What makes data trustworthy

A student types their year level as 91 instead of 9.

The survey only reached the basketball team's group chat.

Rows get duplicated when copied into the spreadsheet.

Ava also collected everyone's favourite colour.

Two hand-counts of the same recess differ by 40.

Read the first letters as a question you ask of any dataset: **A·B·I·R·R** — “Are Biases In Records Real?” It’s clunky on purpose, which is exactly why it sticks. Accuracy, Bias, Integrity, Relevance, Reliability.

Accuracy is not reliability

Students lose marks every year by treating these as synonyms. They’re not. **Accuracy** means data is *close to the true value*. **Reliability** means data is *consistent and repeatable* — you’d get the same result again. A measurement can be one without the other.

Picture darts on a target. Hitting the bullseye every time is **accurate and reliable**. A tight cluster in the top corner is **reliable but not accurate** — consistent, but consistently *wrong* (that’s a systematic error, often *bias*). Darts scattered around the bullseye are **accurate-ish on average but not reliable**. All over the board is neither. Use the buttons to throw each pattern.

“Ava’s hand-tally of courtyard users was **reliable** — she always counted the same way — but not **accurate**, because she stood too far away and consistently undercounted by about ten. Reliable data can still be systematically wrong.”

Inside data integrity

Integrity means the data has been maintained and kept intact and unaltered across its whole life — from the moment it’s collected, through storage, all the way to the visualisation. If a value gets corrupted, duplicated, or quietly edited along the way, integrity is broken even if every other factor looks fine.

Think of how a courtroom treats evidence: it’s logged, sealed, and tracked so nobody can tamper with it between the crime scene and the trial. Data integrity is the same unbroken chain — from Ava’s survey form, into the spreadsheet, into the chart. One unsealed link and the whole conclusion is doubtful.

Here’s the part that trips people up — and it’s a documented exam weakness. The study design lists its **own set of factors for integrity specifically**, and it is *not* the same as the five quality factors. The integrity sub-factors are:

Spot the quality problem

Each card describes something that went wrong in a PulseCheck-style project. Drag it into the quality factor it most directly damages. Some are deliberately close calls — that’s where the marks live.

Practice questions

Attempt each in your notebook first, then reveal the weak answer, the strong answer and the marking guide. Mark yourself honestly — the dock bottom-right keeps your running total.

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Common traps & misconceptions

Exam trap

△ Exam trap — two lists, don't blend them **Quality of data & information** = accuracy, bias, integrity, relevance, reliability (5). **Integrity of data** = accuracy, authenticity, correctness, reasonableness, relevance, timeliness (6). A past VCAA multiple-choice question hinged on spotting that "completeness" is *not* an integrity factor. Note the two overlaps that examiners exploit: **accuracy** and **relevance** appear in *both* lists.

Exam trap

① Quality factors ≠ integrity factors Five quality factors (A·B·I·R·R) vs six integrity factors (accuracy, authenticity, correctness, reasonableness, relevance, timeliness). "Completeness" belongs to *neither* official list. Keep them in separate boxes in your head.

Exam trap

② Accurate vs reliable Accurate = close to true. Reliable = consistent / repeatable. Always give an example of the gap (a scale that's always 5 kg over is reliable but not accurate) — examples are where the second mark hides.

Exam trap

③ Bias is structural, not "an opinion" Bias means the data is systematically skewed by *how it was collected* — e.g. only sharing the survey in the basketball team chat. Don't define it as "someone being unfair." Name the source of the skew.

Exam trap

④ Don't confuse KK03 with KK04 This dot point is the *quality* of data (trustworthiness). The next one — *characteristics* of data & information (size, structure, accessibility, clarity, context) — is different. If a question says "quality," answer with A·B·I·R·R.

Exam trap

⑤ One word never earns a mark A documented examiner note: criteria or explanations written as a single word score zero. A quality criterion must be a measurable statement — "at least 90% of survey records fall within the valid year-level range (7–12)," not just "accurate."

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Distinguish · 2 marks

Between the **accuracy** and the **reliability** of data, using an example.

✗ A weak answer

Accuracy and reliability both mean the data is good and can be trusted.

✓ A strong answer

Accuracy is how close a value is to the true value, whereas reliability is how consistent and repeatable the data is. For example, a sensor that always reads 2 °C above the real temperature is reliable (consistent) but not accurate (it's wrong every time).

How the marks are awarded

- 1 mark — correct distinction (accuracy = closeness to true value; reliability = consistency/repeatability).
- 1 mark — a relevant example that shows the gap between the two.

Q2. Describe · 3 marks

Ava promoted her survey only on the senior basketball group chat. Describe how this could affect the quality of her data, and describe one technique to improve it.

✗ A weak answer

It's biased because not many people answered, so the data isn't fair.

✓ A strong answer

The data would suffer from sampling **bias**: only students already interested in ball sports saw the survey, so the responses systematically over-represent that group and don't reflect the whole student body. This skews the conclusion toward sport. Ava could improve it by distributing the survey through a whole-school channel (e.g. the daily notices or homeroom), giving every student an equal chance to respond.

How the marks are awarded

- 1 mark — identifies bias (specifically sampling bias).
- 1 mark — explains the effect: unrepresentative sample skewed/misleading conclusion.
- 1 mark — a valid improvement that addresses the skew (broaden the distribution).

Q3. State · 2 marks

A response records a student's weekly courtyard time as 200 hours. State which factor of data integrity this breaches, and name two other factors that affect the integrity of data.

✗ A weak answer

It breaches reliability. Other factors are bias and relevance.

✓ A strong answer

It breaches **reasonableness** — 200 hours in a week is not believable (a week only has 168). Two other integrity factors are **accuracy** and **timeliness** (others acceptable: authenticity, correctness, relevance).

How the marks are awarded

- 1 mark — identifies reasonableness as the breached factor.
- 1 mark — names two valid integrity factors from: accuracy, authenticity, correctness, reasonableness, relevance, timeliness. (Bias and reliability are quality factors, not integrity factors — no mark.)

Q4. Explain · 4 marks

Between collecting her survey and building her chart, Ava copied the data into a spreadsheet by hand and accidentally duplicated 30 rows. Explain how this affects the integrity of her data and the quality of the information she presents to the SRC.

✗ A weak answer

The data is wrong now because there are extra rows, so the chart is wrong too.

✓ A strong answer

The duplication breaks **integrity**: the data was altered after collection, so it no longer faithfully represents what students submitted (the chain from collection to visualisation is no longer intact). Because counts and any averages are now inflated by the repeated rows, the derived **information** is inaccurate — the chart over-states demand for whichever option those 30 rows favoured. The SRC could then make a decision based on a conclusion that isn't real. To restore integrity, Ava should verify the row count against the source and remove duplicates before visualising.

How the marks are awarded

- 1 mark — identifies integrity is broken (data altered after collection).
- 1 mark — links it to the collectionvisualisation chain no longer being intact.
- 1 mark — explains the effect on the information (inflated counts/averages misleading chart).
- 1 mark — consequence or remedy (poor SRC decision / verify & de-duplicate).